Catching connected cars that lie about where they are

RMIT University

A signed message proves which car sent it, but it cannot prove that the position inside is true. In our simulations one receiver cannot detect a consistent lie, and receivers that pool their radio measurements catch most lies larger than about **47 m** (95 percent interval 40.6 to 57.3 m).

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1 The problem

Connected cars broadcast their position several times a second over a direct radio link, and the cars around them act on it. Every message is signed, so a car with valid credentials can still sign a false position, and no cryptographic check will reject it.

A receiver does have one independent clue: **how strongly the signal arrived**, which falls with distance from where the sender really is.

2 Aims and approach

- **Can one receiver tell** that a correctly signed position is false?
- **Can many receivers tell** by pooling what they measured, and from what size of lie?
- **Where would an informed attacker lie**, and what stops it?
- **Can roadside units learn the detector together** without sharing raw data?

We simulated a highway in the ns-3 network simulator [2, 3] with five traffic scenarios, ten kinds of attack, and honest cars that carry realistic position error. The result is a public dataset of **7,916,708 labelled one-second windows** [1]. Part A of this project reported a perfect score on a different dataset; an audit found 96.39 percent of its test rows were also in training, so the data was rebuilt and checked for leakage before any model was trained.

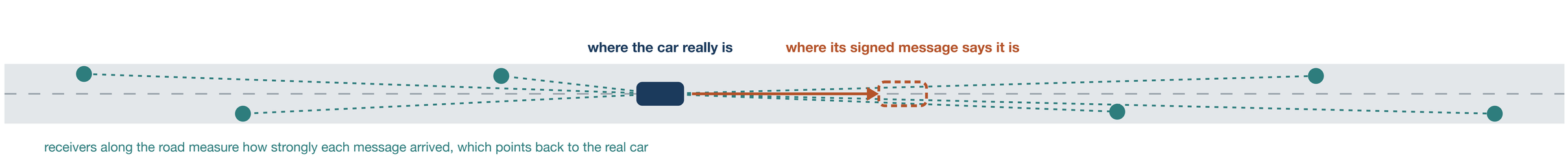


Figure 1. The attack studied here. The message is correctly signed, so only the physics of the radio signal can contradict the claim.

3 Result: a detection floor

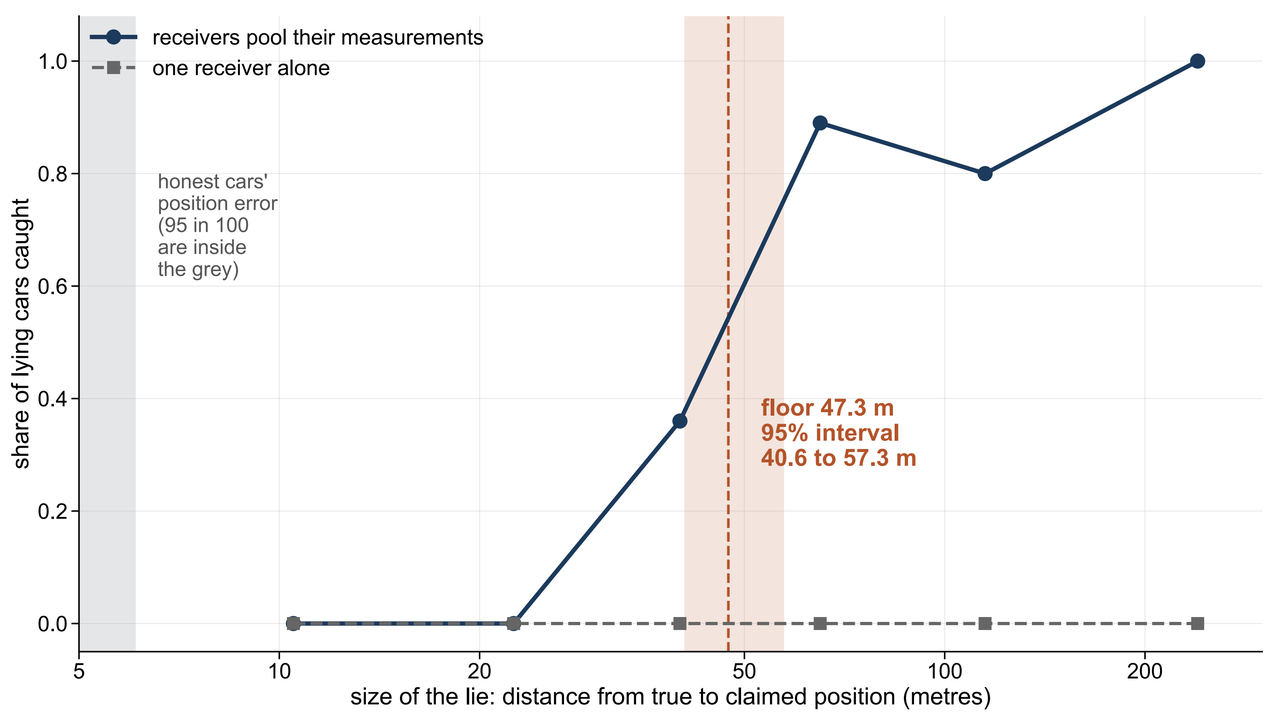


Figure 2. Pooling turns an undetectable lie into a detectable one above about two and a half times the 18.3 m localisation error. Below the floor a lie is too small to separate from honest position error; a lie that small does not move a car into another lane. Counting only lies that stay on the road, which a map check would let through, the floor is 43.4 m, inside the same interval. If each roadside unit pools only the cars near it, the floor is 55.3 m.

One receiver sees nothing. A car that shifts every claim by the same amount keeps its speed and heading consistent. On a detection score from 0 (none caught) to 1 (all caught), content checks score 0.000 on it and four different learning methods reach at most 0.158.

> **0.000**
> detection score of content checks on a consistent lie

Pooling has to happen before the decision. When receivers vote with their verdicts instead of sharing measurements, detection of the same lies stays at exactly 0.000.

> **47.3 m**
> detection floor, 95 percent interval 40.6 to 57.3 m

The honest cars matter. Here they carry realistic position error. Without it every lie is separable in principle and the detection task becomes far easier than on a real road.

4 Result: the geometry has a weak side, and a map closes it

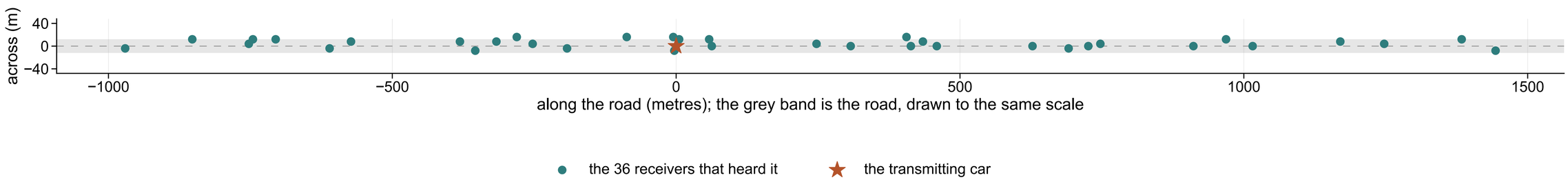


Figure 3. The receivers that heard one car span 2,414 m along the road and 24 m across it. They measure distance along the road well and across it poorly, so an attacker who knows this lies sideways.



Figure 4. Receiver positions alone predict the attacker's best direction: 79.7° off the road against 75 to 85° found by search, for the estimator this system runs.

Table 1. Share of lies caught when the attacker chooses its claim to evade the radio check, at 5 percent false alarms. The map check rejects claims more than 18 m from the road centre, which no honest car exceeds here.

Size of lie	Free to leave the road	Map check at 18 m
25 m	0.002	**0.006**
50 m	0.013	**0.230**
100 m	0.056	**0.854**
200 m	0.241	**0.943**

The two checks cover each other. A sideways lie leaves the road and the map rejects it; a lie along the road is the one the radio sees best.

5 Conclusions

1. **Aim 1.** A single receiver cannot detect a consistent position lie at any size in this data (Figure 2, flat line). The same holds on the public VeReMi datasets [4].
2. **Aim 2.** Pooling received power across receivers detects lies above a floor of 47.3 m, 95 percent interval 40.6 to 57.3 m (Figure 2), or 55.3 m (45.0 to 69.2 m) when each roadside unit pools only its own area. Sharing verdicts instead of measurements detects none.
3. **Aim 3.** Receiver geometry predicts where an informed attacker lies, for the estimator this system runs (Figures 3 and 4). A map check covers that direction (Table 1).
4. **Aim 4.** Trained across roadside units (12 per road, eight simulated runs) without sharing raw data, federated learning comes within about 5 percent of central training on cars no unit has heard, after 200 rounds. Moved into congested traffic, a detector recovers 88 percent of its loss by refitting input scaling on unlabelled local data and fine-tuning briefly.

6 Limits and next step

All measurements are simulated, on one straight highway, and the radio model has not yet been checked against real sidelink measurements. **Next: fit the same propagation law to a public set of real measurements**, then test the floor on a junction and a curve, where the receivers no longer sit in a line.

Try it yourself: you are the attacker. Move a claimed position and watch the receivers decide.

References

1. L. L. Gowda, CV2X-IDS dataset, version 1.1.0, RMIT University, 2026, CC BY 4.0.
2. G. F. Riley and T. R. Henderson, "The ns-3 network simulator," Springer, 2010.
3. N. Patriciello et al., "An E2E simulator for 5G NR networks," Simul. Model. Pract. Theory, 2019.
4. R. W. van der Heijden, T. Lukaseder and F. Kargl, "VeReMi: A dataset for comparable evaluation of misbehavior detection," SecureComm, 2018.